Chapter 5 · Solutions

P24 · Count pairs, then check feasibility

Hint. Total means each entity has at least one partner. Many-to-one adds an upper bound on A's partners.

In many-to-many, at least 3 pairs cover the 3 A entities. The pairs (a1,b1),(a2,b1),(a3,b2) also cover both B entities, attaining the bound. All six possible A×B pairs attain the maximum 6.

In many-to-one, each A has at most one B. Total participation gives at least one, so exactly one. There are exactly 3 pairs. The same minimum example covers both B entities.

With only one A, there can be only one pair, but both B entities require a partner. The conditions are infeasible. Do not report a numerical minimum for an impossible instance.

P25 · Keep each fact once

Hint. Identify independent objects, repeated attributes and relationship facts separately.

Table	Primary key	Reference or purpose
Department(DeptID,Name)	DeptID	one department
Student(StudentID,Name,DeptID)	StudentID	DeptID references Department; NOT NULL
Course(CourseID,Title)	CourseID	one course
Phone(StudentID,Phone)	StudentID,Phone	owner references Student
Enrolment(StudentID,CourseID)	StudentID,CourseID	references both entities
Dependent(StudentID,DependentName)	StudentID,DependentName	owner references Student

There are 6 tables under the stated facts. DependentName alone need not be unique across students. A nullable Phone column in Student can represent at most one phone, so the Phone table can disappear in (b). The count becomes 5.

Check. A several-phone attribute cannot be stored as a repeating group in the original atomic-value Student schema.

P26 · A changed upper bound changes the mapping

Hint. A many-to-many pair is an independent fact; a many-to-one link can be a reference.

The four tables are E1, E2, E2Value(E2ID,Value), and Link(E1ID,E2ID). The multivalued table preserves independent values; the link table preserves all permitted pairs. Both-side total participation does not remove either need.

With many-to-one from E1 to E2, put a NOT NULL E2ID reference in E1 and remove Link. The minimum becomes 3. E2ID need not be unique in E1 because several E1 entities may share one E2.